

SPG-BANDIT: SKILL PROFILE GUIDED BANDIT FOR TASK SELECTION IN LLM AGENT SKILL EVOLUTION

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1 INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Large language model (LLM) agents have demonstrated competence on complex tasks in diverse environments, including embodied control, web navigation, and software engineering. A key mechanism supporting this breadth of capability is *skill evolution*, through which an agent reflects on execution trajectories, extracts reusable knowledge, and accumulates it in a skill library for later use.

Existing work has primarily focused on two components of this process. The first is *skill quality*, namely the extraction of informative and reusable knowledge from trajectories. For example, SkillRL uses reflection to identify reusable strategies from successful episodes and recurrent failure modes from unsuccessful episodes. The second is *skill utilization*, namely the organization, retrieval, and incorporation of skills into an agent’s inference-time prompt. Representative approaches include structured skill banks with category-specific retrieval, vector database retrieval, and dynamic prompting.

A critical dimension remains comparatively underexplored: **the order in which tasks are selected for practice**. Existing systems commonly cycle uniformly through a fixed task pool, sample tasks at random, or rely on hand-designed curricula that do not adapt to an agent’s evolving competence. Since tasks can make substantially different contributions to skill progress, an adaptive selection policy may use the interaction budget more efficiently. This motivates the following question: *Can task selection conditioned on an agent’s current skill profile improve the efficiency and effectiveness of skill evolution in LLM agents?*

1.2 CONTRIBUTIONS

We propose **SPG-Bandit** (Skill Profile Guided Bandit), a framework that casts task selection as a contextual bandit problem informed by an agent’s multidimensional skill profile.

The central idea is to use Multidimensional Item Response Theory (MIRT) to estimate latent proficiency across K skill dimensions and then select tasks expected to improve dimensions with lower estimated proficiency.

Specifically, our contributions are threefold:

1. **Problem formulation.** We formalize task selection during LLM agent skill evolution as a nonstationary contextual bandit problem. The formulation captures both task specific differences in expected skill progress and the multidimensional nature of agent competence.
2. **Method.** We propose SPG-Bandit, which combines MIRT based latent skill profiling with a gap weighted sliding window UCB policy. The method maintains an online proficiency estimate and selects tasks with high predicted progress on underdeveloped dimensions while accounting for estimation uncertainty.
3. **General framework.** We provide a modular framework that can be integrated with existing skill-evolution methods and supports systematic evaluation of task-selection efficiency and downstream task performance.

2 PROBLEM FORMULATION

2.1 NOTATION AND SETUP

Consider an LLM agent interacting with a fixed task pool $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_M\}$ containing M tasks. At round $t = 1, 2, \dots$, the agent selects and executes one task $\tau_t \in \mathcal{T}$ and observes a binary outcome $y_t \in \{0, 1\}$, where one denotes success and zero denotes failure.

The agent has an unobserved *true latent skill profile* $\mathbf{s}_t^* \in [0, 1]^K$, which represents its proficiency across K skill dimensions. The algorithm instead uses the MIRT posterior estimate $\hat{\mathbf{s}}_t \in [0, 1]^K$, which is measurable with respect to the history available before round t . The true skill change is $\Delta_t^* = \mathbf{s}_{t+1}^* - \mathbf{s}_t^*$. Because it is unobserved, the bandit receives the surrogate progress observation $\hat{\delta}_t = \hat{\mathbf{s}}_{t+1} - \hat{\mathbf{s}}_t$. Thus, the method optimizes estimated progress, which is the quantity available from the MIRT filter, rather than directly optimizing Δ_t^* .

Each task τ has a *task embedding* $\mathbf{e}_\tau \in \mathbb{R}^{d_e}$, where $d_e \in \mathbb{N}_+$ is the embedding dimension. The embedding can be obtained by applying a sentence encoder to the natural language description of the task.

2.2 PROBLEM DEFINITION

We formalize adaptive task selection as a *nonstationary contextual bandit* problem. At each round t :

1. The agent observes its context: the estimated profile $\hat{\mathbf{s}}_t$ and the embedding $\mathbf{e}_\tau$ of every task $\tau \in \mathcal{T}$.
2. The agent selects an arm $\tau_t \in \mathcal{T}$.
3. The agent executes τ_t , observes the binary outcome y_t , and updates its skills through reflection.
4. The true profile evolves to $\mathbf{s}_{t+1}^*$, and MIRT produces the next estimate $\hat{\mathbf{s}}_{t+1}$.

To prioritize dimensions with lower estimated proficiency, define the *gap weight* vector

$$\hat{\mathbf{g}}_t = \text{softmax}\left(\frac{\mathbf{1} - \hat{\mathbf{s}}_t}{\gamma_g}\right),$$

where $\mathbf{1} \in \mathbb{R}^K$ is the all ones vector and $\gamma_g > 0$ is a temperature parameter. We use gap weighted skill progress as the immediate bandit reward,

$$\hat{r}_t = \hat{\mathbf{g}}_t^\top \hat{\delta}_t.$$

The objective is to maximize the expected cumulative reward over a bandit horizon of T rounds,

$$\max_{\pi} \mathbb{E} \left[\sum_{t=1}^T \hat{r}_t \mid \tau_t \sim \pi(\cdot \mid \hat{\mathbf{s}}_t, \{\mathbf{e}_\tau\}) \right].$$

where π denotes the task-selection policy and T is the number of rounds in the bandit phase. This reward prioritizes progress on underdeveloped dimensions without requiring an auxiliary objective.

We learn a predictor $\hat{\delta}_t(\tau)$ of the surrogate progress and use $\hat{\mathbf{g}}_t^\top \hat{\delta}_t(\tau)$ as the estimated bandit reward. A superscript $\star$ denotes an unobserved true quantity or an oracle regression parameter, whereas a hat denotes an estimate computed from data. Fixed task embeddings and algorithmic tuning parameters carry neither marker.

3 METHOD

3.1 OVERVIEW

SPG-Bandit consists of a warmup phase and a bandit phase. During **warmup**, the agent samples tasks uniformly to obtain initial interaction outcomes. A sequential MIRT model estimates the latent profile sequence $\{\hat{\mathbf{s}}_t\}$, the surrogate progress vectors $\{\hat{\delta}_t\}$, and task item parameters. These estimates supervise an MLP that maps a task embedding and the current estimated profile to a representation suitable for predicting surrogate progress.

During the **bandit phase**, the method computes $\hat{\mathbf{g}}_t$ from $\hat{\mathbf{s}}_t$ and evaluates every candidate task using a sliding window neural linear UCB rule. The MLP supplies a fixed representation of the observed estimated context, while the ridge head estimates gap weighted surrogate progress and its uncertainty. After execution, an online MIRT update produces $\hat{\mathbf{s}}_{t+1}$, from which $\hat{\delta}_t$ is computed. The tuple $(\phi_{t,\tau_t}, \hat{\delta}_t)$ is then added to the sliding window.

3.2 KEY COMPONENTS

Multidimensional IRT (MIRT). The MIRT data generating model associates each task τ with an unobserved discrimination vector $\mathbf{a}_\tau^* \in \mathbb{R}^K$ and an unobserved difficulty $d_\tau^* \in \mathbb{R}$. Given $\mathbf{s}_t^*$, its success probability is

$$p_\tau^*(\mathbf{s}_t^*) = \sigma(\mathbf{a}_\tau^{*\top} \mathbf{s}_t^* - d_\tau^*).$$

where $\sigma(x) = 1/(1 + e^{-x})$. The estimator constrains $\hat{\mathbf{a}}_\tau \geq 0$ elementwise and $\hat{d}_\tau \geq 0$. The first constraint ensures that increasing an estimated skill component cannot reduce the fitted success probability. These constraints provide a directionally interpretable skill scale.

After collecting N warmup outcomes $\{(\tau_t, y_t)\}_{t=1}^N$, we estimate profiles and item parameters by alternating optimization. The estimator uses $p(\hat{\mathbf{s}}_{t+1} \mid \hat{\mathbf{s}}_t) = \mathcal{N}_{[0,1]^K}(\hat{\mathbf{s}}_t, \mathbf{\Sigma}_s)$, where $\mathbf{\Sigma}_s \succ 0$ is a fixed temporal covariance matrix and $\mathcal{N}_{[0,1]^K}$ denotes a Gaussian distribution truncated to $[0, 1]^K$. This prior regularizes adjacent estimates and stabilizes inference.

E-step. With $\{(\hat{\mathbf{a}}_\tau, \hat{d}_\tau)\}$ fixed, compute $\hat{\mathbf{s}}_{1:N+1}$ by maximizing the joint log posterior

$$\log p(\hat{\mathbf{s}}_1) + \sum_{t=1}^N \left[\log p(y_t \mid \hat{\mathbf{s}}_t, \hat{\mathbf{a}}_{\tau_t}, \hat{d}_{\tau_t}) + \log p(\hat{\mathbf{s}}_{t+1} \mid \hat{\mathbf{s}}_t) \right],$$

where $p(\mathbf{s}_1)$ is a truncated Gaussian prior on $[0, 1]^K$ centered at 0.5.

M-step. With $\{\hat{\mathbf{s}}_t\}$ fixed, update $(\hat{\mathbf{a}}_\tau, \hat{d}_\tau)$ using L-BFGS-B, subject to $\hat{\mathbf{a}}_\tau \geq 0$, $\hat{d}_\tau \geq 0$, and ℓ_2 regularization on $\hat{\mathbf{a}}_\tau$.

We alternate the E-step and M-step until the objective converges.

Online update. During the bandit phase, each observation (y_t, τ_t) refines the profile estimate without rerunning the full alternating procedure:

$$\hat{\mathbf{s}}_{t+1} = \arg \max_{\mathbf{s} \in [0,1]^K} \left\{ \log p(y_t \mid \mathbf{s}, \hat{\mathbf{a}}_{\tau_t}, \hat{d}_{\tau_t}) + \log p(\mathbf{s} \mid \hat{\mathbf{s}}_t) \right\}.$$

Delta prediction via MLP. To predict surrogate progress, we train a two layer MLP $\hat{\Phi}$. Let $\text{cat}(\mathbf{u}, \mathbf{v})$ denote vector concatenation. The learned encoder maps $\text{cat}(\mathbf{e}_\tau, \hat{\mathbf{s}})$ to $\phi \in \mathbb{R}^{d_f}$. Its parameters are estimates and are denoted by $\hat{\mathbf{W}}_1, \hat{\mathbf{b}}_1, \hat{\mathbf{W}}_2, \hat{\mathbf{b}}_2$.

$$\begin{aligned} \mathbf{h} &= \max\left(\hat{\mathbf{W}}_1 \text{cat}(\mathbf{e}_\tau, \hat{\mathbf{s}}) + \hat{\mathbf{b}}_1, 0\right), \\ \phi &= \hat{\mathbf{W}}_2 \mathbf{h} + \hat{\mathbf{b}}_2. \end{aligned}$$

A linear pretraining head $\widehat{\mathbf{W}}_3 \in \mathbb{R}^{d_f \times K}$ predicts surrogate progress as $\widehat{\boldsymbol{\delta}} = \widehat{\mathbf{W}}_3^\top \boldsymbol{\phi}$. The encoder is trained on $\{(\text{cat}(\mathbf{e}_{\tau_i}, \widehat{\mathbf{s}}_i), \widehat{\boldsymbol{\delta}}_i)\}_{i=1}^N$. After pretraining, $\widehat{\mathbf{W}}_3$ is discarded and $\widehat{\Phi}$ is fixed.

Gap-weighted UCB selection. To accommodate a non stationary context to progress relationship, we maintain a sliding window containing the ℓ most recent *completed* interactions and fit a local ridge head within that window. Let $\mathcal{W}_t$ denote the indices stored in the buffer immediately before bandit round t . The buffer is initialized with the final ℓ warmup observations. For each buffered observation i , define $\phi_i = \widehat{\Phi}(\text{cat}(\mathbf{e}_{\tau_i}, \widehat{\mathbf{s}}_i))$. With ridge regularization parameter $\lambda > 0$, the covariance and response matrices are

$$\widehat{\mathbf{W}}_t = \mathbf{A}_t^{-1} \mathbf{B}_t, \quad \mathbf{A}_t = \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top, \quad \mathbf{B}_t = \sum_{i \in \mathcal{W}_t} \phi_i \widehat{\boldsymbol{\delta}}_i^\top,$$

For a candidate task τ , let $\phi_{t,\tau} = \widehat{\Phi}(\text{cat}(\mathbf{e}_\tau, \widehat{\mathbf{s}}_t))$. The predicted surrogate progress is $\widehat{\boldsymbol{\delta}}_t(\tau) = \widehat{\mathbf{W}}_t^\top \phi_{t,\tau}$. The UCB score is

$$\text{score}_t(\tau) = \widehat{\mathbf{g}}_t^\top \widehat{\boldsymbol{\delta}}_t(\tau) + \beta_t \|\widehat{\mathbf{g}}_t\|_2 \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}},$$

The first term favors tasks with high predicted gap weighted progress. The second term is a confidence bonus for the scalarized reward $\widehat{\mathbf{g}}_t^\top \widehat{\boldsymbol{\delta}}_t$, where $\beta_t > 0$ is selected according to a ridge regression confidence bound.

Sliding-window update. After observing $\boldsymbol{\delta}_t$, we append $(\phi_{t,\tau_t}, \boldsymbol{\delta}_t)$ to the window and evict its oldest entry when the window exceeds size ℓ . The incremental updates are:

$$\begin{aligned} \mathbf{A}_{t+1} &= \mathbf{A}_t + \phi_{t,\tau_t} \phi_{t,\tau_t}^\top - \phi_{t-\ell} \phi_{t-\ell}^\top, \\ \mathbf{B}_{t+1} &= \mathbf{B}_t + \phi_{t,\tau_t} \boldsymbol{\delta}_t^\top - \phi_{t-\ell} \boldsymbol{\delta}_{t-\ell}^\top, \\ \widehat{\mathbf{W}}_{t+1} &= \mathbf{A}_{t+1}^{-1} \mathbf{B}_{t+1}. \end{aligned}$$

The window size ℓ controls the tradeoff between estimation quality, which benefits from a larger sample, and responsiveness to nonstationarity, which benefits from recent observations.

3.3 ALGORITHM

Algorithm 1 summarizes the complete procedure.

4 THEORETICAL ANALYSIS

Because the relationship between context and skill progress may change over time, expected task rewards can be non stationary. In this section, $t \in \{1, \dots, T\}$ indexes only bandit phase rounds. We analyze the dynamic *pseudo regret* of SPG-Bandit under the sliding window ridge estimator described in Section 3.

4.1 NON-STATIONARY SETTING

For each candidate task τ , define $\phi_{t,\tau} = \widehat{\Phi}(\text{cat}(\mathbf{e}_\tau, \widehat{\mathbf{s}}_t))$ and the conditional mean surrogate progress

$$\boldsymbol{\mu}_t(\tau) = \mathbb{E}[\widehat{\boldsymbol{\delta}}_t(\tau) \mid \mathcal{F}_{t-1}].$$

The expected scalar reward is $q_t(\tau) = \widehat{\mathbf{g}}_t^\top \boldsymbol{\mu}_t(\tau)$. Let $\tau_t^* \in \arg \max_{\tau \in \mathcal{T}} q_t(\tau)$ be the dynamic oracle action. We evaluate the dynamic pseudoregret

$$R_T = \sum_{t=1}^T [q_t(\tau_t^*) - q_t(\tau_t)].$$

Algorithm 1 SPG-Bandit

Input: Task pool $\mathcal{T}$, warmup steps $N \geq \ell$, bandit horizon T , window size ℓ , feature dimension d_f , gap temperature γ_g , confidence radii $\{\beta_t\}$, regularizer $\lambda > 0$, skill dimensions K

Phase 1: Warmup

for $i = 1$ to N **do**

Select $\tau_i \in \mathcal{T}$ uniformly

Execute τ_i , observe $y_i \in \{0, 1\}$, and perform the skill-evolution update

Store (τ_i, y_i)

end for

Run sequential MIRT EM to obtain $\{\hat{\mathbf{s}}_i\}_{i=1}^{N+1}$, $\{\hat{\boldsymbol{\delta}}_i\}_{i=1}^N$, and $\{\hat{\mathbf{a}}_\tau, \hat{d}_\tau\}_{\tau \in \mathcal{T}}$

Train and freeze $\hat{\Phi}$ on $\{(\text{cat}(\mathbf{e}_{\tau_i}, \hat{\mathbf{s}}_i), \hat{\boldsymbol{\delta}}_i)\}_{i=1}^N$

Reindex the final ℓ warmup tuples as $\{(\tau_i, \mathbf{s}_i, \phi_i, \boldsymbol{\delta}_i)\}_{i=1-\ell}^0$, where $\phi_i = \Phi(\text{cat}(\mathbf{e}_{\tau_i}, \mathbf{s}_i))$

Set $\mathcal{W}_1 = \{1 - \ell, \dots, 0\}$, $\mathbf{A}_1 = \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_1} \phi_i \phi_i^\top$, $\mathbf{B}_1 = \sum_{i \in \mathcal{W}_1} \phi_i \hat{\boldsymbol{\delta}}_i^\top$, $\hat{\mathbf{W}}_1 = \mathbf{A}_1^{-1} \mathbf{B}_1$, and $\hat{\mathbf{s}}_1 = \hat{\mathbf{s}}_{N+1}$

Phase 2: Bandit

for $t = 1$ to T **do**

Compute $\hat{\mathbf{g}}_t = \text{softmax}((\mathbf{1} - \hat{\mathbf{s}}_t)/\gamma_g)$

for each $\tau \in \mathcal{T}$ **do**

$\phi_{t,\tau} = \hat{\Phi}(\text{cat}(\mathbf{e}_\tau, \hat{\mathbf{s}}_t))$, $\hat{\boldsymbol{\delta}}_t(\tau) = \hat{\mathbf{W}}_t^\top \phi_{t,\tau}$

$\text{score}_t(\tau) = \hat{\mathbf{g}}_t^\top \hat{\boldsymbol{\delta}}_t(\tau) + \beta_t \|\hat{\mathbf{g}}_t\|_2 \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}$

end for

$\tau_t = \arg \max_{\tau \in \mathcal{T}} \text{score}_t(\tau)$

Execute τ_t and observe y_t

$\hat{\mathbf{s}}_{t+1} \leftarrow \text{BAYESUPDATE}(\hat{\mathbf{s}}_t, \hat{\mathbf{a}}_{\tau_t}, \hat{d}_{\tau_t}, y_t)$

$\hat{\boldsymbol{\delta}}_t \leftarrow \hat{\mathbf{s}}_{t+1} - \hat{\mathbf{s}}_t$ and $\phi_t \leftarrow \phi_{t,\tau_t}$

$\mathcal{W}_{t+1} \leftarrow (\mathcal{W}_t \cup \{t\}) \setminus \{t - \ell\}$

$\mathbf{A}_{t+1} \leftarrow \mathbf{A}_t + \phi_t \phi_t^\top - \phi_{t-\ell} \phi_{t-\ell}^\top$

$\mathbf{B}_{t+1} \leftarrow \mathbf{B}_t + \phi_t \hat{\boldsymbol{\delta}}_t^\top - \phi_{t-\ell} \hat{\boldsymbol{\delta}}_{t-\ell}^\top$

$\hat{\mathbf{W}}_{t+1} \leftarrow \mathbf{A}_{t+1}^{-1} \mathbf{B}_{t+1}$

end for

This quantity is a pseudo regret because it compares conditional expected rewards rather than realized noisy outcomes. Let $\mathcal{F}_{t-1}$ denote the history available before round t . The estimated context $\hat{\mathbf{s}}_t$, and hence $\hat{\mathbf{g}}_t$ and $\phi_{t,\tau}$, are $\mathcal{F}_{t-1}$ -measurable.

4.2 ASSUMPTIONS

1. **(Time varying linear surrogate model.)** There are oracle matrices $\{\mathbf{W}_t^*\}_{t=1}^T$, where $\mathbf{W}_t^* \in \mathbb{R}^{d_f \times K}$, such that $\hat{\boldsymbol{\delta}}_t(\tau) = \mathbf{W}_t^{*\top} \phi_{t,\tau} + \varepsilon_t(\tau)$, with $\mathbb{E}[\varepsilon_t(\tau) \mid \mathcal{F}_{t-1}, \tau] = 0$.
2. **(Boundedness and noise.)** $\|\phi_{t,\tau}\|_2 \leq 1$, $\|\mathbf{W}_t^*\|_F \leq S$, and, for every $\mathbf{u}$ satisfying $\|\mathbf{u}\|_2 \leq 1$, the scalar $\mathbf{u}^\top \varepsilon_t(\tau)$ is conditionally σ -sub-Gaussian.
3. **(Parameter path variation.)** The reward model variation is $P_T = \sum_{t=1}^{T-1} \|\mathbf{W}_{t+1}^* - \mathbf{W}_t^*\|_F$.
4. **(Calibrated window and coverage.)** Before the first bandit round, warmup provides ℓ calibration observations generated under $\mathbf{W}_1^*$. At each bandit round, the window contains exactly ℓ observations and satisfies $\sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top \succeq \kappa \ell \mathbf{I}_{d_f}$ for a constant $\kappa > 0$.

Assumption 1 matches the algorithmic regression target, which is $\boldsymbol{\delta}_t$ rather than $\mathbf{s}_{t+1}$. Assumption 2 controls the scale of features, parameters, and noise. Assumption 3 quantifies non stationarity through the parameter path. Assumption 4 isolates the effect of non stationarity from an ill conditioned design matrix. For a failure probability $\rho \in (0, 1)$, Appendix A provides a detailed high probability proof.

4.3 DYNAMIC REGRET DECOMPOSITION

Let $w_t(\tau) = \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}$ and $\widehat{\mathbf{W}}_t = \mathbf{A}_t^{-1} \mathbf{B}_t$. Under Assumptions 1 to 4, Appendix A proves that, with probability at least $1 - \rho$,

$$\left| \mathbf{g}_t^\top (\widehat{\mathbf{W}}_t - \mathbf{W}_t^*)^\top \phi_{t,\tau} \right| \leq \beta_t \|\mathbf{g}_t\|_2 w_t(\tau) + D_t,$$

where $\beta_t = \tilde{O}(\sigma\sqrt{Kd_f} + \sqrt{\lambda}S)$ is a confidence radius and D_t is the explicitly bounded bias from stale observations. Since the policy selects the task with the largest UCB score, approximate optimism yields

$$q_t(\tau_t^*) - q_t(\tau_t) \leq 2\beta_t \|\mathbf{g}_t\|_2 w_t(\tau_t) + 2D_t.$$

Because $\mathbf{g}_t$ is a softmax vector, $\|\mathbf{g}_t\|_2 \leq 1$. Assumption 4 gives $w_t(\tau) \leq (\lambda + \kappa\ell)^{-1/2}$ for every candidate task. The detailed counting argument in Appendix A gives $\sum_{t=1}^T D_t \leq \ell^2 P_T / \sqrt{\lambda(\lambda + \kappa\ell)}$. Therefore, with $\beta = \max_{t \leq T} \beta_t$,

$$R_T \leq \underbrace{\frac{2\beta T}{\sqrt{\lambda + \kappa\ell}}}_{\text{estimation}} + \underbrace{\frac{2\ell^2 P_T}{\sqrt{\lambda(\lambda + \kappa\ell)}}}_{\text{drift}}.$$

4.4 OPTIMAL WINDOW AND REGRET BOUND

For $P_T > 0$, when $\kappa\ell \geq \lambda$, balancing the two terms yields the following feasible integer window size:

$$\ell^* = \min \left\{ T, \max \left\{ \left\lceil \frac{\lambda}{\kappa} \right\rceil, \left\lceil \left(\frac{\beta T \sqrt{\lambda}}{P_T} \right)^{1/2} \right\rceil \right\} \right\},$$

and the corresponding cumulative regret bound, up to universal constants:

$$R_T = O \left(\frac{(\beta T)^{3/4} P_T^{1/4}}{\kappa^{1/2} \lambda^{1/8}} \right).$$

If $P_T = 0$, the drift term vanishes and $\ell = T$ minimizes the displayed bound. If $P_T = o(T)$ and β , κ^{-1} , and λ^{-1} grow at most polylogarithmically, then the bound is sublinear. For example, if $P_T = O(T^\nu)$ for $\nu < 1$, then

$$R_T = \tilde{O} \left(T^{(3+\nu)/4} \right) = o(T).$$

A DETAILED DYNAMIC PSEUDO-REGRET ANALYSIS

This appendix proves the dynamic pseudo regret bound in Section 4. Every bound below is conditional on Assumptions 1 to 4. In particular, the coverage assumption is essential because the stated result does not hold for an arbitrary sequence of feature vectors.

A.1 NOTATION

Table A1. Notation used in the method and theoretical analysis.

Symbol	Definition
t, i, j, T, N, M , and K	Bandit-round index, calibration index, auxiliary index, bandit horizon, warmup length, number of tasks, and number of skill dimensions.
$\mathcal{T} = \{\tau_1, \dots, \tau_M\}$, τ_t , and τ_t^*	Task pool, task selected by the policy, and dynamic oracle task at round t .
$y_t \in \{0, 1\}$	Binary outcome observed after executing τ_t .
$\mathbf{s}_t^*$ and $\widehat{\mathbf{s}}_t \in [0, 1]^K$	True latent skill profile and its MIRT posterior estimate.
Δ_t^* and $\widehat{\delta}_t$	True skill progress and observed surrogate progress, respectively.
$\mathbf{a}_\tau^*$, d_τ^* and $\widehat{\mathbf{a}}_\tau$, $\widehat{d}_\tau$	True MIRT item parameters and their estimates.
Σ_s and $\mathcal{N}_{[0,1]^K}$	Temporal covariance matrix and Gaussian distribution truncated to $[0, 1]^K$.
$\mathbf{e}_\tau \in \mathbb{R}^{d_e}$, d_e, d_h, d_f	Fixed task embedding and the embedding, hidden, and feature dimensions.
$\widehat{\Phi}$, $\widehat{\mathbf{W}}_1$, $\widehat{\mathbf{b}}_1$, $\widehat{\mathbf{W}}_2$, $\widehat{\mathbf{b}}_2$, $\widehat{\mathbf{W}}_3$	Learned MLP encoder, its learned layer parameters, and its discarded pre-training head.
$\text{cat}(\cdot, \cdot)$, $\mathbf{h}$, and $\phi_{t,\tau} \in \mathbb{R}^{d_f}$	Vector concatenation, MLP hidden activation, and feature computed from $\mathbf{e}_\tau$ and $\widehat{\mathbf{s}}_t$.
γ_g , $\mathbf{1}$, and $\widehat{\mathbf{g}}_t$	Gap temperature, K -dimensional all-ones vector, and estimated softmax gap weight vector.
$\widehat{r}_t$ and $\widehat{\delta}_t(\tau)$	Observed scalar surrogate reward and predicted surrogate progress for candidate task τ .
$\mathbf{W}_t^* \in \mathbb{R}^{d_f \times K}$ and $\widehat{\mathbf{W}}_t$	Oracle time-varying linear surrogate parameter and its sliding-window ridge estimate.
$\epsilon_t(\tau)$, σ , and S	Regression noise, its conditional sub-Gaussian scale, and an upper bound on $\ \mathbf{W}_t^*\ _F$.
$\mathcal{F}_{t-1}$, $\mathcal{W}_t$, and ℓ	History before round t , sliding-window index set, and fixed window size.
$\mathbf{A}_t$, $\mathbf{B}_t$, and $\lambda > 0$	Ridge covariance matrix, response matrix, and regularization parameter.
$w_t(\tau)$, $\mathbf{Z}_t$, and $\mathbf{e}_k$	Elliptical feature width, windowed noise matrix, and the k -th Euclidean basis vector.
β_t , ρ , and κ	Confidence radius, failure probability, and window-coverage constant.
P_T , v_j , V_t^{win} , and D_t	Parameter-path variation, one-step variation, window variation, and stale-window drift allowance.
$\mu_t(\tau)$, $q_t(\tau)$, $U_t(\tau)$, and R_T	Conditional mean surrogate progress, expected scalar reward, UCB width, and dynamic pseudoregret.

A.2 INDEXING AND FILTRATION

The bandit phase is indexed by $t \in \{1, \dots, T\}$. To avoid a boundary case in the first ℓ rounds, we regard the ℓ calibration observations from warmup as rounds $1 - \ell, \dots, 0$. By Assumption 4, they are generated with parameter $\mathbf{W}_1^*$. For every bandit round t , define

$$\mathcal{W}_t = \{t - \ell, \dots, t - 1\}.$$

Thus, $|\mathcal{W}_t| = \ell$ for every t . This indexing is equivalent to the buffer implementation in Algorithm 1.

Let $\mathcal{F}_{t-1}$ be the sigma-field generated by the calibration observations and all selected tasks and observed outcomes before bandit round t . For an observed task at index i , abbreviate $\phi_i = \phi_{i,\tau_i}$. Assumption 1 gives

$$\delta_i = \mathbf{W}_i^{*\top} \phi_i + \epsilon_i, \quad \mathbb{E}[\epsilon_i \mid \mathcal{F}_{i-1}, \tau_i] = \mathbf{0}. \quad (1)$$

For calibration indices $i \leq 0$, set $\mathbf{W}_i^* = \mathbf{W}_1^*$. Consequently, calibration data do not contribute to the parameter-path variation.

At the beginning of round t , ridge regression uses

$$\begin{aligned} \mathbf{A}_t &= \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top, \\ \mathbf{B}_t &= \sum_{i \in \mathcal{W}_t} \phi_i \delta_i^\top, \\ \widehat{\mathbf{W}}_t &= \mathbf{A}_t^{-1} \mathbf{B}_t. \end{aligned} \quad (2)$$

For a candidate task τ , write

$$w_t(\tau) = \|\phi_{t,\tau}\|_{\mathbf{A}_t^{-1}} = \sqrt{\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_{t,\tau}}.$$

A.3 A UNIFORM RIDGE-REGRESSION CONFIDENCE BOUND

For a failure probability $\rho \in (0, 1)$, define

$$\beta_t = \sigma \sqrt{2K \log \left(\frac{KT \det(\mathbf{A}_t)^{1/2}}{\rho \lambda^{d_f/2}} \right)} + \sqrt{\lambda} S.$$

Let

$$\mathbf{Z}_t = \sum_{i \in \mathcal{W}_t} \phi_i \varepsilon_i^\top.$$

For the k -th output coordinate and a fixed window $\mathcal{W}_t$, the self-normalized martingale inequality gives

$$\left\| \mathbf{A}_t^{-1/2} \mathbf{Z}_t \mathbf{e}_k \right\|_2 \leq \sigma \sqrt{2 \log \left(\frac{KT \det(\mathbf{A}_t)^{1/2}}{\rho \lambda^{d_f/2}} \right)}.$$

Applying a union bound over the K output coordinates and the T windows makes the inequality hold simultaneously for every $k \in \{1, \dots, K\}$ and $t \leq T$, with probability at least $1 - \rho$. The factor KT inside the logarithm records this union bound. Summing the squared coordinate bounds yields

$$\left\| \mathbf{A}_t^{-1/2} \mathbf{Z}_t \right\|_F \leq \sigma \sqrt{2K \log \left(\frac{KT \det(\mathbf{A}_t)^{1/2}}{\rho \lambda^{d_f/2}} \right)}. \quad (3)$$

A.4 PREDICTION ERROR DECOMPOSITION

Substituting equation 1 into equation 2 gives

$$\mathbf{B}_t = \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top \mathbf{W}_i^* + \mathbf{Z}_t.$$

Since $\mathbf{A}_t = \lambda \mathbf{I}_{d_f} + \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top$, adding and subtracting $\mathbf{W}_t^*$ gives

$$\begin{aligned} \widehat{\mathbf{W}}_t - \mathbf{W}_t^* &= \mathbf{A}_t^{-1} \mathbf{Z}_t - \lambda \mathbf{A}_t^{-1} \mathbf{W}_t^* \\ &\quad + \mathbf{A}_t^{-1} \sum_{i \in \mathcal{W}_t} \phi_i \phi_i^\top (\mathbf{W}_i^* - \mathbf{W}_t^*). \end{aligned} \quad (4)$$

The three terms are the stochastic estimation term, the regularization term, and the stale-window bias.

Fix a candidate task τ . Cauchy Schwarz and equation 3 give

$$|\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \mathbf{Z}_t \mathbf{g}_t| \leq w_t(\tau) \left\| \mathbf{A}_t^{-1/2} \mathbf{Z}_t \right\|_F \|\mathbf{g}_t\|_2.$$

Moreover, since $\mathbf{A}_t \succeq \lambda \mathbf{I}_{d_f}$,

$$|\lambda \phi_{t,\tau}^\top \mathbf{A}_t^{-1} \mathbf{W}_t^* \mathbf{g}_t| \leq \sqrt{\lambda} S w_t(\tau) \|\mathbf{g}_t\|_2.$$

For the stale-window bias, define

$$V_t^{\text{win}} = \sum_{i \in \mathcal{W}_t} \|\mathbf{W}_i^* - \mathbf{W}_t^*\|_F.$$

For every $i \in \mathcal{W}_t$, Cauchy Schwarz and $\mathbf{A}_t \succeq \lambda \mathbf{I}_{d_f}$ imply

$$\begin{aligned} |\phi_{t,\tau}^\top \mathbf{A}_t^{-1} \phi_i| &\leq \left\| \mathbf{A}_t^{-1/2} \phi_{t,\tau} \right\|_2 \left\| \mathbf{A}_t^{-1/2} \phi_i \right\|_2 \\ &\leq \frac{w_t(\tau)}{\sqrt{\lambda}}. \end{aligned}$$

Using $\|\phi_i\|_2 \leq 1$, the bias term in equation 4 is bounded by $\frac{w_t(\tau) \|\mathbf{g}_t\|_2}{\sqrt{\lambda}} V_t^{\text{win}}$. Assumption 4 gives $\mathbf{A}_t \succeq (\lambda + \kappa \ell) \mathbf{I}_{d_f}$, and therefore

$$w_t(\tau) \leq \frac{1}{\sqrt{\lambda + \kappa \ell}}. \quad (5)$$

Define

$$D_t = \frac{V_t^{\text{win}}}{\sqrt{\lambda(\lambda + \kappa\ell)}}. \quad (6)$$

Combining equation 4 with equation 6 proves that, simultaneously for all $t \leq T$ and $\tau \in \mathcal{T}$,

$$\left| \mathbf{g}_t^\top (\widehat{\mathbf{W}}_t - \mathbf{W}_t^*)^\top \phi_{t,\tau} \right| \leq \beta_t \|\mathbf{g}_t\|_2 w_t(\tau) + D_t. \quad (7)$$

A.5 SUMMING THE DRIFT ALLOWANCES

Let $v_j = \|\mathbf{W}_{j+1}^* - \mathbf{W}_j^*\|_F$ for $j \geq 1$, and set $v_j = 0$ for $j \leq 0$. The triangle inequality gives, for $i < t$,

$$\|\mathbf{W}_i^* - \mathbf{W}_t^*\|_F \leq \sum_{j=i}^{t-1} v_j.$$

Thus,

$$\sum_{t=1}^T V_t^{\text{win}} \leq \sum_{t=1}^T \sum_{i=t-\ell}^{t-1} \sum_{j=i}^{t-1} v_j \quad (8)$$

$$\leq \ell^2 \sum_{j=1}^{T-1} v_j = \ell^2 P_T. \quad (9)$$

For each fixed increment v_j , there are at most ℓ windows that contain it and at most ℓ indices within each such window. Combining equation 6 and equation 9 gives

$$\sum_{t=1}^T D_t \leq \frac{\ell^2 P_T}{\sqrt{\lambda(\lambda + \kappa\ell)}}. \quad (10)$$

A.6 FROM CONFIDENCE TO DYNAMIC PSEUDOREGRET

Define $U_t(\tau) = \beta_t \|\mathbf{g}_t\|_2 w_t(\tau)$. Equation 7 gives, for every candidate task,

$$q_t(\tau) \leq \mathbf{g}_t^\top \widehat{\mathbf{W}}_t^\top \phi_{t,\tau} + U_t(\tau) + D_t.$$

The policy selects τ_t to maximize the first two terms on the right-hand side. Therefore,

$$q_t(\tau_t^*) \leq \mathbf{g}_t^\top \widehat{\mathbf{W}}_t^\top \phi_{t,\tau_t} + U_t(\tau_t) + D_t \quad (11)$$

$$\leq q_t(\tau_t) + 2U_t(\tau_t) + 2D_t. \quad (12)$$

The second inequality follows by applying the lower counterpart of equation 7 to the selected task.

Since $\mathbf{g}_t$ is a softmax vector, $\|\mathbf{g}_t\|_2 \leq 1$. Equation 5 gives

$$\sum_{t=1}^T U_t(\tau_t) \leq \frac{\beta T}{\sqrt{\lambda + \kappa\ell}}, \quad \beta = \max_{t \leq T} \beta_t. \quad (13)$$

Summing equation 12, then applying equation 10 and equation 13, establishes

$$R_T \leq \frac{2\beta T}{\sqrt{\lambda + \kappa\ell}} + \frac{2\ell^2 P_T}{\sqrt{\lambda(\lambda + \kappa\ell)}}. \quad (14)$$

When $\kappa\ell \geq \lambda$, the two terms in equation 14 have orders $\beta T/(\sqrt{\kappa\ell})$ and $\ell^{3/2} P_T/(\sqrt{\kappa\lambda})$. For $P_T > 0$, equating these orders gives $\ell = (\beta T \sqrt{\lambda}/P_T)^{1/2}$. Rounding upward, enforcing $\kappa\ell \geq \lambda$, and truncating at T yields the window size in Section 4. If $P_T = 0$, the drift term is zero and the largest feasible window is optimal. $\square$